

CARE-DS: Critical-Neighborhood-Aware Residual-Energy Dominating-Set Scheduling for Heterogeneous Wireless Sensor Networks

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ABSTRACT In a duty-cycled wireless sensor network, an active dominating set provides service while other sensors sleep. Existing methods commonly precompute a pairwise-disjoint family. Such a schedule can repeat a set while every member is affordable, but it may strand energy when one member becomes the bottleneck and cannot construct overlapping replacements. We study dynamic scheduling in which each round may use a new dominating set or sink-rooted connected dominating set, subject to residual energy and an explicit handover cost. We derive a closed-neighborhood resource bound for uniform activation cost and use its residual-energy analogue as a decision score to design critical-neighborhood-aware residual-energy dominating-set scheduling (CARE-DS). The algorithm constructs a portfolio of feasible sets, ranks them by minimum post-decision neighborhood reserve, and uses normalized dwell control to avoid unnecessary handovers. The evaluation contains 2700 paired instances from five topology families, 30 seeds, 50–500 vertices, three densities, four handover costs, and two service models. At handover costs 0.25 and 0.5, CARE-DS improves aggregate mean lifetime over a retrospective per-instance best dynamic comparator by 2.88–2.81% for ordinary domination and 3.28–3.48% for connected domination; the differences remain significant after Holm correction. We also evaluate modern static heuristics up to 1,000 vertices, graph-valid inter-sensor first-order radio routing, controlled failures, ablations, and exact sparse ordinary-domination instances. An explicitly exploratory, within-stratum lower-tail analysis shows that the nonzero-cost radio gains rise to 31.54% and 5.59% on the lowest best-comparator-lifetime decile under ordinary and connected service, respectively. CARE-DS is not best when handovers are free; its advantage occurs when reconfiguration consumes energy or the surviving-node set changes.

INDEX TERMS Connected dominating set, dominating set, handover cost, network lifetime, residual energy, sleep–wake scheduling, surviving-network continuity, wireless sensor networks.

I. INTRODUCTION

Wireless sensor networks (WSNs) are used when measurements must be collected over a region without keeping every sensor awake. Examples include environmental and agricultural monitoring, industrial inspection, structural monitoring, and smart-city sensing. These applications differ in scale and traffic pattern, but they share one restriction: sensor batteries are difficult to replace, whereas sensing, communication, and state changes consume energy [1]–[4]. Network lifetime therefore depends not only on the energy of individual sensors, but also on how sensing and forwarding duties are distributed over time.

Let the sensors be the vertices of a communication graph. Two sensors are adjacent when they can communicate directly. A set D of active sensors is a dominating set if every sensor

outside D has a neighbor in D . Thus, a dominating set gives a simple graph model for one round of local service: the active sensors sense, receive, or aggregate data, while the other sensors may sleep. If the subgraph induced by D is connected and contains the sink, then D also gives a virtual backbone through which the collected data can be forwarded. We study both interpretations because local coverage alone and coverage with sink reachability are distinct requirements [5], [6].

The usual scheduling approach is to compute pairwise-disjoint dominating sets and activate them one after another [7]. Disjointness is attractive because a sensor then belongs to at most one set. It is, however, a construction rule and not a service requirement. Consider a sensor with enough energy for ten rounds that is placed in a set whose other members can serve for only two rounds. A fixed disjoint family may

leave most of its energy unused. The restriction is more serious when the energy distribution is heterogeneous, when a few neighborhoods have very little total energy, or when a failed sensor invalidates one of the precomputed sets.

Changing the active set is not free either. A newly activated node may have to wake its radio, synchronize with its neighbors, receive a schedule, and join a collection route [8], [9]. A method that reconstructs a good set in every round may therefore lose energy through the very changes that were intended to extend its lifetime. This gives two competing objectives. The schedule should move duties away from critically depleted neighborhoods, but it should not change its membership without sufficient benefit.

We consider this dynamic scheduling problem. In every round, the algorithm may select any affordable dominating set, or sink-rooted connected dominating set, from the surviving sensors. The same sensor may be used in several nonconsecutive rounds. The energy of each selected sensor and the cost of newly entering the active set are deducted before the next decision. Service ends at the first round for which no affordable feasible set exists. The principal question is whether a scheduler can protect the locally scarce energy that every future dominating set must use while avoiding excessive handovers.

Our answer is critical-neighborhood-aware residual-energy dominating-set scheduling (CARE-DS). Every future dominating set must meet each closed neighborhood. This gives a resource upper bound on lifetime and, more importantly, an online measure of which part of the network is closest to losing service. CARE-DS constructs several feasible candidate sets and prefers the set that leaves the largest minimum neighborhood reserve after the current round. A dwell rule retains a feasible set when the handover cost makes an immediate change wasteful.

The main contributions are as follows.

- 1) We formulate energy-budgeted dynamic dominating-set scheduling with optional sink-rooted connectivity, explicit handover cost, failures, and a precise continued-service termination rule.
- 2) We derive a closed-neighborhood resource upper bound for the unit-cost model and use a resource-inspired residual score as the primary online decision signal. Under nonuniform radio expenditure, the score is explicitly treated as a heuristic rather than a lifetime bound.
- 3) We propose CARE-DS, which combines a diverse candidate portfolio, reserve-aware selection, and cost-aware dwell control. We give complete pseudocode and establish feasibility of every reported service round.
- 4) We perform a paired evaluation over five graph families, including an exploratory stratified lower-tail robustness analysis, a shared-portfolio dynamic control with measured runtime, first-order radio energy, controlled failures, component ablations, and exact cases.
- 5) We compare with validated reproductions of recent static methods and evaluate FSS both on its native unweighted objective and on heterogeneous WSN lifetime.

TABLE 1. Graph objects and their operational interpretation.

Graph object	WSN interpretation
Vertex v	Battery-powered sensor with residual energy b_v
Edge uv	Direct communication link available in the current topology
Dominating set	Active sensors that locally serve every live sensor
Connected dominating set	Active sink-rooted coverage and forwarding backbone
Set entry	Wake-up, synchronization, schedule, and route-joining cost
$N[v]$ reserve	Energy locally available to serve v in future rounds

The result is deliberately conditional. CARE-DS is not best when handovers are free. It becomes useful when changing the active set consumes energy and when failures force the schedule to be reconstructed. This is the operating regime for which the method is proposed.

The rest of the paper is organized as follows. Section II describes the application setting and the resulting design requirements. Section III places the work among static and application-oriented methods. Sections IV–VI give the model, resource bound, and CARE-DS algorithm. The experimental design and results follow, after which we discuss the application-level meaning and the limitations of the evidence.

II. APPLICATION SETTING AND DESIGN REQUIREMENTS

We use a round-based duty-cycle model. At the beginning of a refresh event, a coordinator obtains the live-node topology and the available energy of each sensor. It then selects an active set, announces the schedule, and constructs a collection route. During the service round, active nodes sense, receive, relay, or aggregate data according to the selected service model. The measured energy cost is deducted, failed nodes are removed, and the process is repeated until no affordable active set can provide the required service.

Table 1 gives the correspondence between the graph model and the network operation. The ordinary domination model is appropriate when an active sensor can locally serve a sleeping neighbor and a separate mechanism carries the aggregated data to the sink. The connected model is stricter: the active sensors themselves must contain a sink-rooted forwarding backbone. Their comparison estimates the additional lifetime restriction of the connected-service model under the stated sink and routing assumptions; it does not isolate a single causal routing penalty.

The application gives four design requirements. First, *coverage must be checked in every round*; an algorithm cannot count residual energy after a sensor has ceased to receive service. Second, *residual energy must remain reusable*; pairwise disjointness should not prevent a high-energy sensor from returning in a later round. Third, *membership changes must be charged*; otherwise a rapidly changing schedule receives an

unrealistic advantage. Fourth, *the schedule must recover from failures*; both the selectable sensors and the sensors requiring service change when a node is removed.

The study assumes that the coordinator has exact state information at refresh events. This corresponds to a centralized upper layer rather than a fully distributed medium-access-control (MAC) protocol. The assumption lets us isolate the scheduling question, but it also means that the present experiments do not measure the control packets required to obtain that state. Section XI returns to this limitation.

In this paper, “heterogeneous” refers specifically to unequal initial and residual sensor-energy budgets. Packet size, radio-electronics parameters, communication capability, and sensing demand are homogeneous unless a stated failure changes the live-node universe. The experiments therefore evaluate energy heterogeneity, not heterogeneous radio types or sensing modalities.

III. RELATED WORK AND POSITIONING

The related literature can be viewed in three groups. The first uses disjoint dominating sets as a WSN schedule. The second develops stronger search methods for the maximum disjoint dominating sets problem (MDDSP). The third adds energy-aware set formation and scheduling. These groups are closely related, but they optimize different objects.

Pino et al. start from disjoint dominating sets and improve heterogeneous-energy lifetime by swapping vertices among existing sets [3]. Their work shows why the number of sets alone is insufficient when sensor energies are different. It also leaves open two directions that are important here: allowing the number and membership of the scheduled sets to change, and treating connected dominating sets. Balbal et al. formulate maximum weighted disjoint dominating sets and introduce the Greedy Heuristic for Maximum Weighted Disjoint Dominating Sets (GH-MWDDS) and its plus variant [10]. These methods form the immediate static background of the present work.

The use of connected dominating sets as routing backbones predates the lifetime-oriented disjoint-family methods. Guha and Khuller give approximation algorithms for connected domination [5], and several centralized and distributed constructions have since been studied for ad hoc and sensor networks [6]. Fault-tolerant variants additionally require the backbone to survive node loss [11]. Our connected model does not impose a prescribed level of vertex connectivity; it reconstructs a connected set after the declared failures.

Recent MDDSP work improves the construction of a static disjoint family. The population-based iterated greedy method (PBIG) uses a population, randomized greedy reconstruction, destruction, adaptive parameters, and restart [12]. Multi-Constructor CMSA merges components produced by several Construct, Merge, Solve, and Adapt (CMSA) constructors and solves a reduced exact subproblem [13]. Fixed Set Search (FSS) learns common solution components from an elite set and applies swap and 1-in-2-out local search. Its experiments report strong results on unweighted ER, Watts–Strogatz, and

Barabasi–Albert instances [14]. These are important static comparators. They do not, however, decide which overlapping set should be activated from the current residual-energy state.

The 2024 energy-aware algorithm (EAA), also described as an energy-aware disjoint dominating-set algorithm (EADDSA), forms energy-attentive disjoint dominating sets (DSs) by progressive deletion and schedules them using estimated set lifetimes [4]. Energy-aware disjoint families have also been combined with population-based methods such as whale optimization [15]. Its 2026 extension, denoted EAAS–S4C–MAB by its authors, creates a size/lifetime skyline, classifies sets into four cases, applies upper-confidence-bound (UCB) scheduling with activation caps and cooldowns, and uses chain collection [16]. It is the closest application baseline in this study. Its published model retains a formed disjoint family. CARE-DS instead reconstructs an overlapping candidate portfolio from the current live-node budgets. It also includes explicit handover expenditure, failure-triggered reconstruction, and sink-rooted connected domination.

The distinction can be stated simply. Static MDDSP asks for many pairwise-disjoint feasible sets. The problem studied here asks for a long sequence of affordable feasible sets, where the cost of a set depends on the current energy and on the previously active set. The former is a special case of the latter, as shown in Section IV.

IV. SYSTEM MODEL AND PROBLEM STATEMENT

A. NETWORK AND SERVICE MODEL

Let $G = (V, E)$ be an undirected communication graph, $n = |V|$, and $N[v]$ the closed neighborhood of v . Node v has initial energy $b_v > 0$. In service round t , the active set is $D_t \subseteq V_t$, where V_t is the surviving node set. Ordinary service requires

$$D_t \cap N[v] \neq \emptyset, \quad \forall v \in V_t. \quad (1)$$

Connected service additionally requires a live sink $s \in D_t$ and connectivity of $G[D_t]$. The sink is externally powered in the connected experiments.

B. ENERGY AND TRANSITION COST

Under the unit model, an active non-sink node consumes $a > 0$ and a node newly entering the active set consumes handover energy $h \geq 0$. With binary activity x_{vt} and entry $z_{vt} \geq x_{vt} - x_{v,t-1}$, energy feasibility is

$$\sum_t (ax_{vt} + hz_{vt}) \leq b_v, \quad \forall v \in V. \quad (2)$$

The objective is the number of consecutive feasible service rounds before no affordable set satisfying (1) and optional connectivity exists. This definition prevents an algorithm from claiming additional lifetime after coverage has already failed.

For radio experiments, a k -bit transmission over distance d costs

$$E_{tx}(k, d) = k(E_{elec} + \epsilon_{amp}d^2), \quad (3)$$

reception costs $E_{rx}(k) = kE_{elec}$, and aggregation costs $E_{agg}(k) = kE_{DA}^{bit}$ [17]. Each active ordinary-DS node generates

one k -bit report. Active terminals are ordered by a deterministic nearest-neighbor rule following the Power-Efficient Gathering in Sensor Information Systems (PEGASIS) collection principle [18]. Every consecutive terminal pair is joined by a minimum-hop path in G ; every inter-sensor path hop is therefore a declared communication edge, and all intermediate relay transmissions and receptions are charged. The destination terminal aggregates the arriving packet with its own report without increasing packet length. The last terminal makes one direct transmission to the external sink; this final collector hop is not required to be an edge of the sensor graph. Connected DSs use a sink-rooted breadth-first tree in $G[D_t]$ and charge one transmission per non-sink node and one reception and aggregation per received child packet. Links are bidirectional; sensing, retransmission, contention, and fading energy are omitted. A radio candidate is affordable only when every terminal and intermediate relay can pay its complete cost. Any terminal or relay absent from the preceding working route also pays h . Nodes with depleted energy are excluded from route search. Every algorithm receives the same deterministic routing and cost function. In ordinary radio mode, let $W(D)$ denote the union of the DS terminals and every intermediate relay on these graph paths. In connected mode, $W(D) = D \setminus \{s\}$. Thus $W(D)$ is precisely the positive-cost support of a round, and handover expenditure is charged to $W(D_t) \setminus W(D_{t-1})$.

C. DYNAMIC SCHEDULING PROBLEM

The problem considered in this paper is the following.

Dynamic WSN Dominating-Set Scheduling. Given G , the initial energy vector $\mathbf{b}$, activation and handover costs, a failure sequence, and a choice of ordinary or connected service, find a longest consecutive sequence $(D_1, D_2, \dots, D_T)$ such that every D_t satisfies the service condition on the live graph and the complete per-node energy expenditure never exceeds $\mathbf{b}$.

This definition makes the dependence between consecutive decisions explicit. The affordability of D_t depends on all preceding sets, and its handover cost depends on D_{t-1} . A failure changes both the set of selectable vertices and the set of vertices that must be dominated. Thus, selecting a small dominating set in isolation is not the same as constructing a long schedule.

D. RELATION TO STATIC MDDSP

When $b_v = a$ for every vertex, $h = 0$, and there are no failures, each vertex can appear in at most one round. Maximizing dynamic lifetime is then exactly the maximum disjoint dominating sets problem (MDDSP). The proposed problem is therefore NP-hard, and its additional energy, connectivity, and transition terms strictly generalize the static case.

V. CRITICAL-NEIGHBORHOOD RESERVE BOUND

Every dominating set intersects every $N[v]$. With a uniform activation cost, the nodes in one closed neighborhood can

collectively supply at most their integer activation capacities. Hence

$$T^* \leq U(\mathbf{b}) = \min_{v \in V} \sum_{u \in N[v]} \left\lfloor \frac{b_u}{a} \right\rfloor. \quad (4)$$

Proposition 1. Equation (4) upper-bounds every feasible ordinary-domination schedule. It also bounds connected schedules after neighborhoods already covered by the free sink are excluded.

Proof. Fix a vertex v . Each service round selects at least one energy-consuming vertex in $N[v]$, while those vertices jointly supply no more than the sum in (4) activations. This holds for every v , so the minimum is an upper bound. If the powered sink belongs to $N[v]$, that neighborhood yields no finite energy restriction and is omitted. Connectivity only removes feasible active sets. $\square$

For connected service define $Q_t = \{v \in V_t : s \notin N[v]\}$; for ordinary service let $Q_t = V_t$. For candidate D at the current round, let $\mathbf{b}'(D)$ be the residual budget after activation, radio, and handover costs. CARE-DS maximizes the residual analogue

$$R(D) = \min_{v \in Q_t} \sum_{u \in N[v] \cap V_t} b'_u(D), \quad (5)$$

with $R(D) = +\infty$ if $Q_t = \emptyset$. Thus, the externally powered sink's random sensor-budget field neither enters the minimum nor affects connected candidate ranking.

Proposition 1 and (4) are unit-cost results. Equation (5) is their continuous residual-energy analogue: in the unit model it tracks the local resource that supports the bound. Under radio expenditure, per-round costs depend on route position and relay use, so $R(D)$ is a resource-inspired heuristic score, not a radio-lifetime upper bound. Joule sums remain comparable within each radio experiment because every sensor uses the same packet size and radio parameters; candidate-specific routing expenditure is deducted before the score is evaluated.

VI. CARE-DS

A. OVERVIEW

CARE-DS separates the decision into two parts. The first part constructs a small collection of structurally different feasible sets. The second part asks which of these sets leaves the network in the best state after the next round. This separation is useful because no single greedy construction is reliable on all topology families. It also gives a common candidate portfolio to the shared-portfolio control used in the experiments.

The complete round scheduler is given in Algorithm 1. A portfolio is normally retained for $H = 100$ rounds. It is reconstructed earlier when a failure occurs or when none of its sets remains affordable. If a forced refresh still gives no affordable set, CARE-DS terminates because its incomplete candidate portfolio has been exhausted. This is an algorithmic stopping condition, not a proof that no feasible set exists outside the portfolio.

Algorithm 1: CARE-DS dynamic scheduler

Input: graph G , energy $\mathbf{b}$, service mode, sink s , activation cost a , handover cost h , refresh interval H , failures.

Output: schedule $(D_1, \dots, D_T)$.

1. Set $t \leftarrow 1$, $D_0 \leftarrow \emptyset$, $V_1 \leftarrow V$, consecutive-use counter $\ell \leftarrow 0$, and portfolio $\mathcal{P} \leftarrow \emptyset$.
2. At the beginning of round t , remove failed nodes from V_t , set their energy to zero, and invalidate $\mathcal{P}$ if a failure occurred.
3. If $\mathcal{P}$ is empty or $t \equiv 1 \pmod{H}$, set $\mathcal{P} \leftarrow \text{BuildPortfolio}(G[V_t], \mathbf{b}, \text{mode}, s)$ using Algorithm 2.
4. Let $L = \max\{1, \lceil \gamma h / a_{\text{ref}} \rceil\}$. If $h > 0$, $\ell < L$, and D_{t-1} remains structurally feasible and affordable, set $D_t \leftarrow D_{t-1}$.
5. Otherwise set $D_t \leftarrow \text{SelectByReserve}(\mathcal{P}, \mathbf{b}, D_{t-1})$ using Algorithm 3.
6. If no set was selected, rebuild $\mathcal{P}$ once from the current state and repeat Step 5. If no set is then selected, return $(D_1, \dots, D_{t-1})$.
7. Validate domination, optional sink-rooted connectivity, and affordability; deduct the complete activation, radio, and handover cost. Set $\ell \leftarrow \ell + 1$ if $D_t = D_{t-1}$ and $\ell \leftarrow 1$ otherwise; set $t \leftarrow t + 1$ and go to Step 2.

B. CANDIDATE PORTFOLIO

At a refresh event, CARE-DS first forms the set A of live nodes with enough energy to participate in another round. In connected mode the sink is included and seeds every construction. The portfolio contains two deletion-based sets, four deterministic greedy sets, and four randomized greedy sets. Duplicate or infeasible sets are removed.

The deletion construction starts with all eligible nodes and visits them in nondecreasing priority. A node is removed when domination and the required connectivity are preserved. The two priority vectors are residual energy and residual energy divided by degree plus one. The constructive greedy method selects a node using

$$g_p(v) = |N[v] \setminus C| b_v^p, \quad (6)$$

where C is the currently covered set and $p \in \{0.5, 1, 1.5, 2\}$. In connected mode only vertices adjacent to the current partial solution are eligible. A final deletion pass makes the returned set inclusion-minimal. In a randomized construction, the restricted candidate list contains vertices whose score is at least $g_{\max} - \alpha(g_{\max} - g_{\min})$, and one member is sampled uniformly; $\alpha \in [0.1, 0.45]$.

Algorithm 2: BuildPortfolio

Input: live graph $G[V_t]$, residual energy $\mathbf{b}$, mode, sink s , number of randomized starts $r = 4$.

Output: duplicate-free feasible portfolio $\mathcal{P}$.

1. Let A be the affordable live vertices; add s in connected mode. If A cannot provide the required service, return $\emptyset$.
2. Set $\mathcal{P} \leftarrow \emptyset$.
3. Add $\text{MinimalDeletion}(A, \mathbf{b})$ and $\text{MinimalDeletion}(A, (b_v/(d(v) + 1))_{v \in V_t})$ to $\mathcal{P}$.
4. For each $p \in \{0.5, 1, 1.5, 2\}$, add the deterministic construction using score (6).
5. Repeat r times: sample $p \in \{0.5, 1, 1.5, 2\}$ and $\alpha \in [0.1, 0.45]$, and add the corresponding restricted-candidate construction.
6. Remove duplicates and every set that fails domination, required connectivity, or affordability. Return the remaining portfolio.

C. RESERVE-AWARE SELECTION AND DWELL CONTROL

For a candidate D , let $\mathbf{c}(D \mid D_{t-1})$ be its complete cost vector and let $\mathbf{b}'(D) = \mathbf{b} - \mathbf{c}(D \mid D_{t-1})$. An unaffordable candidate is discarded. Every other candidate is ranked lexicographically by

$$\left(R(D), \min_{u \in W(D)} b'_u(D), -h|W(D) \setminus W(D_{t-1})|, -|W(D)| \right). \quad (7)$$

The first coordinate protects the global coverage bottleneck. The second avoids selecting a set that immediately exhausts one of its working nodes. The last two coordinates prefer fewer working-route entries and a smaller working set when the reserve coordinates tie. Under the unit model, $W(D) = D$ for ordinary service and $W(D) = D \setminus \{s\}$ for connected service.

Algorithm 3: SelectByReserve

Input: portfolio $\mathcal{P}$, residual energy $\mathbf{b}$, previous set D_{t-1} , and previous working set $W(D_{t-1})$.

Output: selected set D_t or $\emptyset$.

1. Set $D^* \leftarrow \emptyset$ and $S^* \leftarrow (-\infty, -\infty, -\infty, -\infty)$.
2. For every $D \in \mathcal{P}$, compute $\mathbf{c}(D \mid D_{t-1})$. Continue to the next candidate if some $c_v > b_v$.
3. Set $\mathbf{b}' \leftarrow \mathbf{b} - \mathbf{c}$ and compute $R(D)$ from (5).
4. Compute the score $S(D)$ in (7). If $S(D) >_{\text{lex}} S^*$, set $D^* \leftarrow D$ and $S^* \leftarrow S(D)$.
5. Return D^* .

When $h > 0$, a feasible current set is retained for $L = \max\{1, \lceil \gamma h / a_{\text{ref}} \rceil\}$ rounds before score-based replacement, where $\gamma = 100$. In the unit model $a_{\text{ref}} = a$; in the radio model $a_{\text{ref}} = kE_{\text{elec}}$, a reference packet-electronics energy scale that deliberately excludes the distance-dependent amplifier term. The dimensionless ratio h/a_{ref} replaces the earlier unit-dependent expression. The rule never prevents an early change when the current set becomes unaffordable or structurally invalid. The switch-awareness ablation removes this rule and the entry-cost score coordinate.

D. FEASIBILITY AND COMPLEXITY

Proposition 2. Every round returned by CARE-DS satisfies domination, optional sink-rooted connectivity, and nonnegative residual energy.

Proof. A candidate is retained only after structural validation and a componentwise comparison of its complete cost vector with residual energy. Deducting that vector preserves nonnegative budgets. Failures invalidate the portfolio; failed vertices are removed from both the selectable and required live-node universes before reconstruction. Induction over rounds proves the claim. $\square$

Let q be portfolio size and $m = |E|$. Straightforward adjacency-list construction has a conservative per-refresh complexity $O(qn(n + m))$. Cached immutable closed neighborhoods, vectorized coverage sums, and incremental deletion counts reduce the observed cost. Selection from a fixed portfolio requires the complete cost and reserve of every candidate. Runtime is therefore measured and reported; it is not excluded from the comparison.

VII. COMPARATORS AND REPRODUCTION POLICY

The primary dynamic comparators are event-refreshed EAA deletion (D-EAA-ER; here ER means event refresh, not Erdős–Rényi), dynamic weighted greedy (DWG), and wall-time-tuned multi-start randomized greedy (MS-RG-WT). Table 2 gives their complete decision rules. D-EAA-ER starts from all affordable live nodes and deletes nodes in nondecreasing residual-energy order while preserving the required service. It retains the result until the regular refresh, a failure, or loss of affordability. DWG reconstructs one set every round using score $|N[v] \setminus C|/b_v$, followed by low-energy-first deletion of redundant members. MS-RG-WT and CARE-DS use the same portfolio constructors and paired random stream, but not necessarily identical realized portfolios after their state trajectories diverge. MS-RG-WT has no dwell rule and selects the candidate with the largest post-decision selected-node bottleneck, breaking ties by smaller set size.

The factor 0.45 was fixed on the separate calibration matrix before the primary 30-seed matrix: it increased MS-RG-WT refresh frequency until its mean runtime was close to CARE-DS. This calibration controls candidate-generation strength without claiming identical wall time on each instance; measured runtimes are therefore retained and reported.

Static comparators are EAA, GH-MWDDS+, PBIG-R, MC-CMSA-R, the 2026 FSS reproduction (FSS-2026-R), and EAAS-S4C-MAB-R. The suffix “-R” is intentional: no public author executable was located for these exact experiments. Each is a documented paper-based reproduction or, for MC-CMSA, a validated reconstruction. PBIG-R includes a population, weighted reconstruction, adaptive destruction/determinism, and restart. MC-CMSA-R uses six constructors, merge, a reduced binary program, and component aging. FSS-2026-R uses an elite archive, similarity-based fixed cores, swap, and 1-in-2-out search. These three anytime methods receive the same 0.5-second construction budget. EAAS-S4C-MAB-R implements progressive energy-ordered deletion, block-nested-loop (BNL) skyline filtering, $\eta = 0.5$ as the skyline size–lifetime balance parameter, four median-defined cases, $\beta = 0.05$ as the UCB exploration coefficient, disclosed caps/cooldowns, rescue, future-utility reward, and chain collection. A conservative EAA-family fallback strengthens this comparator. All returned families are revalidated for domination and disjointness.

VIII. EXPERIMENTAL DESIGN

The primary matrix uses random geometric (RGG), clustered geometric, Erdős–Rényi (ER), Watts–Strogatz (WS), and Barabási–Albert (BA) graphs; $n \in \{50, 100, 150, 200, 250, 500\}$; target average degrees 15, 25, and 35; and 30 paired seeds. Integer initial energy is sampled independently from the discrete uniform distribution on $[20, 100]$, unit activity cost is one, and $h/a \in \{0, 0.1, 0.25, 0.5\}$. Thus the handover sweep is expressed as a dimensionless fraction of one unit activation. Both ordinary and sink-rooted connected domination are tested.

For RGGs, independent points are uniform on the unit square and the radius is the pairwise-distance order statistic producing approximately $n\bar{d}/2$ edges. Clustered geometric graphs draw nodes around centers $(0.24, 0.25)$, $(0.75, 0.27)$, and $(0.48, 0.75)$ with coordinate standard deviation 0.13 and clipping to the unit square, followed by the same radius rule. ER uses $p = \bar{d}/(n-1)$; WS uses the nearest even ring degree and rewiring probability 0.1; BA uses attachment parameter $\max\{1, \text{round}(\bar{d}/2)\}$. Geometric and random positions are used only for radio distance; WS nodes are placed on a circle. A disconnected draw is rejected up to 500 times. The rare remaining case is connected by the minimum-length Euclidean component bridges, which is recorded by the deterministic generator. Vertex 0 is the powered sink in connected mode and otherwise has no privileged role.

The weighted modern-static matrix extends all five families to $n = 1000$ with 30 seeds and the same three degree targets. A separate unweighted validation uses ER, WS, and BA graphs up to 1,000 vertices; with unit energy, lifetime is the number of disjoint sets. The radio study uses 10 seeds, $n = 100, 250, 500$, all five families and densities, energy uniform on $[0.05, 0.20]$ J, 4,000-bit packets, $E_{elec} = 50$ nJ/bit, $\epsilon_{amp} = 100$ pJ/bit/m², $E_{DA}^{bit} = 5$ nJ/bit, and a 100-m square. In ordinary radio mode, the external collector is at $(50, 110)$ m. Connected radio mode instead uses the generated graph vertex 0 at its generated position as an externally powered embedded sink; the external coordinate is not used in that mode. Consequently, ordinary and connected results are service-model evaluations, not a controlled estimate of a single connectivity or sink-distance effect. Wake-up cost is 0 or 10^{-4} J. Since $a_{ref} = kE_{elec} = 2 \times 10^{-4}$ J, the nonzero radio setting has $h/a_{ref} = 0.5$. This value is an order-of-magnitude protocol proxy rather than a universal device constant. For context, a CC2420-class IEEE 802.15.4 radio draws approximately 18.8 mA in receive mode and requires oscillator and regulator startup before synchronization traffic is exchanged [19]; the complete entry transaction can therefore be of the same order as 10^{-4} J, depending on supply voltage and control traffic.

Failures remove 2% of nodes at rounds 25 and 50. Random failures sample live non-sink nodes uniformly. Controlled low-reserve removal selects nodes with the least initial closed-neighborhood energy reserve. The schedule continues only while all surviving nodes receive service; because a removed node is no longer a service target, absolute lifetime is interpreted together with the no-failure control. The ablation removes (i) the reserve coordinate and (ii) both handover-aware dwell and score terms. A fixed, outcome-independent exact stress suite uses sparse 21–22 vertex graphs with energy in $[1, 6]$. A second twelve-instance ordinary-domination challenge suite uses 22 vertices and energy in $[1, 8]$; it is selected from a disjoint calibration pool for low heuristic-to-bound utilization and is labeled adversarial rather than used for population inference.

A one-factor-at-a-time sensitivity study uses 100 additional paired graphs: all five families, $n \in \{100, 250\}$, target degrees 15 and 35, and five seeds. It varies dwell

TABLE 2. Primary dynamic controls. “On exhaustion” means that an immediate reconstruction is attempted when no retained portfolio candidate remains affordable.

Method	Candidate construction	Refresh policy	Selection and handover treatment
D-EAA-ER	One residual-energy deletion set	Every H rounds, after failure, or on exhaustion	Reuse while affordable; otherwise reconstruct; entry energy is charged
DWG	One deterministic residual-energy $\times$ uncovered-neighborhood greedy set	Every round	The single feasible construction is used; entry energy is charged
MS-RG-WT	Same two deletion, four deterministic, and four randomized portfolio constructors as CARE-DS	Every $0.45H$ rounds, after failure, or on exhaustion	Maximize post-decision selected-node bottleneck; smaller-set tie break; no dwell rule
CARE-DS	The declared shared portfolio constructors	Every H rounds, after failure, or on exhaustion	Lexicographic score (7) plus the cost-normalized dwell rule

scale $\gamma \in \{25, 50, 100, 200\}$, regular refresh interval $H \in \{25, 50, 100, 200\}$, and randomized starts $r \in \{0, 4, 8\}$ while holding the other two controls at their declared baseline values. Both service modes and $h \in \{0.25, 0.5\}$ are evaluated.

A separate algorithm-seed robustness subset uses an outcome-independent graph pool. Its dynamic part contains 40 graphs: all five families, $n \in \{100, 500\}$, target degrees 15 and 35, and two graph seeds. CARE-DS and MS-RG-WT are each repeated with ten paired algorithm seeds in both service modes at $h \in \{0.25, 0.5\}$. Its static part contains 36 unweighted ER, WS, and BA graphs, $n \in \{100, 500, 1000\}$, the same two degree targets and two graph seeds; PBIG-R, MC-CMSA-R, and FSS-2026-R are each repeated ten times. Inference for the dynamic subset first averages algorithm repeats within a graph and then uses the 40 graph-level paired differences, so repetitions are not treated as independent graph observations.

The reported exact cases use ordinary domination and set $h = 0$. We enumerate every inclusion-minimal feasible DS and solve the resulting integer capacitated scheduling model: one nonnegative integer variable gives the number of activations of each set, node-incidence constraints enforce the energy budgets, and the objective maximizes total activations. SciPy’s mixed-integer linear-programming interface is run with a 120-s limit. This is a mixed-integer linear program (MILP). An instance is called exact only when the solver returns an optimality status; unresolved instances are not converted into heuristic “optima.” The reported challenge cases all returned the optimal status.

All comparisons are paired by graph and seed. CARE-DS comparisons use two-sided Wilcoxon signed-rank tests and 3,000-sample bootstrap confidence intervals for mean differences. Holm correction is applied separately to two predeclared hypothesis families within each study: all CARE-versus-individual-comparator condition tests, and all CARE-versus-retrospective-best condition tests [20]. The four repeated-seed dynamic cells form a third family and are Holm-adjusted together. Modern-static methods are compared using mean ranks, a Friedman test, and Holm-adjusted two-sided paired Wilcoxon tests. Raw rows include seed, topology, realized degree, lifetime, runtime, switches, activations, mean set size, and validity. Here a condition is one fixed combination of service mode, handover cost, failure scenario, and energy

model over the declared graph instances.

After completion of the primary matrices, we formulated an exploratory lower-tail question: does CARE-DS provide a larger advantage on graph instances for which the competing methods terminate unusually early? For the unit-cost study we retain the material-handover cells $h/a \in \{0.25, 0.5\}$; for the radio study we retain $h/a_{ref} = 0.5$. Within every fixed topology, n , target degree, service mode, and handover-cost stratum, instances are ordered by the lifetime of the per-instance best observed comparator. CARE-DS lifetime is not used to define the ordering. Deterministic instance ID breaks comparator ties. Decile 1 contains the lowest comparator lifetimes and decile 10 the highest. Tail gains are ratios of aggregate means. Their 95% intervals use 3,000 stratified paired-bootstrap replicates: graph instances are resampled within each design stratum and the tail is reselected in every replicate. Because this hypothesis and its cutoffs were specified after the matrices were inspected, we report it as exploratory robustness evidence, not as a preregistered confirmatory test; no additional hypothesis-test p -values are assigned.

Experiments ran on Linux with an AMD EPYC 9V74 processor using Python 3.12.13, NumPy 2.5.2, and SciPy 1.18.0. “Switches” denotes the symmetric-difference membership count $|D_t \Delta D_{t-1}|$ accumulated over consecutive rounds; it is not merely the number of rounds in which any change occurred.

IX. RESULTS

A. INTEGRITY AND PRIMARY LIFETIME RESULTS

All six submission-scale tables passed automated integrity checks defined before final aggregation: 86,400 primary dynamic rows, 18,900 weighted-static rows, 9,450 unweighted-static rows, 9,000 radio rows, 12,150 failure rows, and 8,100 ablation rows. There were no duplicate keys, missing values, or invalid schedules.

Table 3 compares CARE-DS with a deliberately strict retrospective oracle baseline. This is an analysis device, not an executable online method: on each graph instance, it takes the largest lifetime among D-EAA-ER, DWG, and MS-RG-WT. CARE-DS is not best when changing the active set is free. At $h = 0.25$ and 0.5 , however, its aggregate gain over that per-instance oracle is 2.81–2.88% for ordinary domination

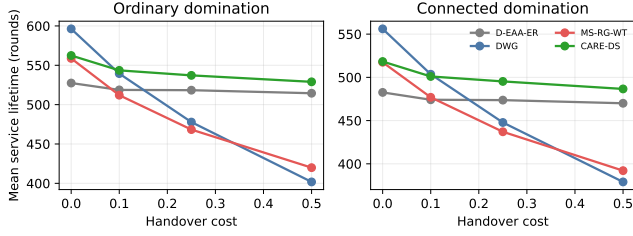


FIGURE 1. Mean lifetime across 2,700 paired instances per service-mode and handover-cost cell.

and 3.28–3.48% for connected domination. All four positive cells remain significant after Holm correction (largest adjusted $p < 3.2 \times 10^{-115}$); the corresponding bootstrap intervals for the mean lifetime difference exclude zero. Table 4 reports uncertainty, effect size, and win/tie/loss counts rather than relying on significance alone.

The global average hides topology dependence. At $h = 0.5$, ordinary per-instance-oracle gains range from 0.44% on clustered graphs to 5.62% on ER graphs. Connected gains are 5.26% on ER and 6.77% on WS, but the mean per-instance percentage is -0.92% on BA and -0.15% on clustered graphs. Thus the evidence supports a handover-aware operating regime, not a claim of universal dominance.

B. EXPLORATORY LOWER-TAIL ROBUSTNESS

The headline averages also hide a stronger low-lifetime effect. We rank graph instances only by best-comparator lifetime within each fixed design stratum; CARE-DS outcomes do not determine tail membership. Table 5 shows that the unit-cost gain increases consistently in the lowest-lifetime quintile and decile. In the radio study the concentration is much stronger: ordinary-service gain rises from 5.34% overall to 31.54% in the lowest decile, where the best comparator averages 136.96 rounds and CARE-DS averages 180.16. Connected radio gain rises from 1.59% to 5.59%.

The stratified-bootstrap interval for the lowest-decile gain excludes zero in all four rows. More importantly, the 95% interval for the contrast between the lowest decile and the remaining instances is also positive: [1.21, 1.86] percentage points for unit-cost ordinary service, [1.68, 2.60] for unit-cost connected service, [15.47, 33.50] for radio ordinary service, and [0.98, 5.17] for radio connected service. Figure 2 reports all ten deciles rather than only the selected cutoffs. The unit-cost trend decreases from the lowest to highest decile in both service modes. Radio ordinary service shows the strongest concentration in the first three deciles; it is not monotone across the higher deciles.

C. COMPUTE-BUDGET CONTROL AND UPPER-BOUND UTILIZATION

MS-RG-WT uses the same portfolio constructors and stochastic stream as CARE-DS but replaces neighborhood reserve by selected-node bottleneck scoring. Its mean wall time is 1.110 s per condition versus 1.169 s for CARE-DS; CARE

is slower for ordinary domination (0.538 versus 0.405 s) but slightly faster for connected domination (1.799 versus 1.814 s). Against this control, CARE improves mean lifetime by 14.7% and 26.0% at $h = 0.25$ and 0.5 for ordinary domination, and by 13.3% and 24.1% for connected domination. The result is therefore not attributable to a larger candidate portfolio or greater mean runtime in this implementation. Deterministic D-EAA-ER and DWG terminate before the available budget; artificially repeating an identical deterministic construction would not enlarge their search.

The closed-neighborhood resource bound is finite on every ordinary instance and on 2,699 of 2,700 connected instances. At $h = 0.25$ and 0.5, CARE uses 83.7% and 82.7% of the ordinary bound, compared with 73.1% and 61.9% for DWG. Connected utilization is 67.8% and 66.8%, compared with 60.9% and 52.1% for DWG. Because the bound omits transition cost and connectivity, it is intentionally optimistic rather than a claimed optimality gap.

D. MODERN STATIC COMPARATORS

The weighted and unweighted studies lead to different rankings (Table 6). Under heterogeneous energy, EAA-static and the strengthened EAAS reproduction tie at mean lifetime 418.97 and mean rank 2.02; the conservative EAA fallback is selected on every weighted instance. PBIG-R is next, while FSS-2026-R and MC-CMSA-R are disadvantaged because maximizing set count does not protect weighted set bottlenecks.

On the native unweighted MDDSP objective, FSS-2026-R is strongest: mean 10.70 sets, mean rank 1.76, best or tied on 1,768 of 1,890 instances, and only 0.74% mean gap to the per-instance best. PBIG-R ranks second. This recovers the qualitative ordering reported by FSS and provides a mechanism-level plausibility check for the reproduction; it does not establish implementation equivalence. It also shows why a static unweighted victory cannot be transferred directly to dynamic heterogeneous lifetime.

E. RADIO ENERGY AND THE CLOSEST APPLICATION BASELINE

Under graph-valid first-order radio routing with no wake-up expenditure, CARE is 5.01% below the per-instance oracle for ordinary domination and 3.30% below it for connected domination. With a 10^{-4} -J wake-up cost, the ordinary gain is 5.34% (95% CI for the paired mean difference [5.35, 18.83] rounds, $r_{tb} = 0.205$, W/T/L=261/14/175, Holm-adjusted $p = 4.1 \times 10^{-4}$); the connected gain is 1.59% (CI [9.86, 25.92], $r_{tb} = 0.156$, W/T/L=236/13/201, adjusted $p = 4.7 \times 10^{-3}$). The corresponding mean per-instance improvements are 12.79% and 1.23%, showing why aggregate-mean and per-instance summaries are reported separately. EAAS-S4C-MAB-R obtains 178.20 ordinary rounds with zero wake-up cost versus 237.46 for CARE; its fixed disjoint family can repeat a feasible set but cannot reconstruct overlapping replacements after a bottleneck. This is a direct implementation-level comparison, not a claim about the authors' unavailable executable.

TABLE 3. CARE-DS versus the per-instance best observed dynamic comparator. $\delta = 100(\bar{T}_{\text{CARE}} / \bar{T}_{\text{Oracle}} - 1)$ is the percentage difference between aggregate means.

h	Ordinary domination			Connected domination		
	CARE	Oracle	Δ	CARE	Oracle	Δ
0	562.43	596.66	-5.74%	518.30	556.72	-6.90%
0.10	543.55	552.31	-1.59%	501.04	512.08	-2.16%
0.25	537.18	522.13	+2.88%	495.31	479.58	+3.28%
0.50	528.99	514.52	+2.81%	486.61	470.26	+3.48%

TABLE 4. Paired evidence for the four primary positive cells. CI is the 3,000-sample bootstrap 95% interval for $T_{\text{CARE}} - T_{\text{Oracle}}$; r_{rb} is the matched-pairs rank-biserial correlation.

Mode	h	Mean difference	95% CI	r_{rb}	W/T/L	Holm p
Ordinary	0.25	15.05	[14.09, 15.96]	0.751	1719/630/351	4.2×10^{-192}
Ordinary	0.50	14.46	[13.43, 15.41]	0.649	1807/434/459	3.3×10^{-157}
Connected	0.25	15.73	[14.54, 16.91]	0.611	1577/542/581	5.7×10^{-133}
Connected	0.50	16.35	[15.10, 17.60]	0.560	1582/483/635	3.1×10^{-115}

TABLE 5. Exploratory lower-tail robustness against the per-instance best observed comparator. Entries are aggregate-mean CARE-DS gains. CI is the stratified-bootstrap 95% interval for the lowest-decile gain. W/T/L is counted within that decile.

Energy study	Service	All	Lowest quintile	Lowest decile (95% CI)	Decile W/T/L
Unit, $h/a \in \{0.25, 0.5\}$	Ordinary	2.85%	4.00%	4.47% [3.95, 4.56]	325/179/36
Unit, $h/a \in \{0.25, 0.5\}$	Connected	3.38%	4.91%	5.51% [4.92, 5.77]	308/185/47
Radio, $h/a_{\text{ref}} = 0.5$	Ordinary	5.34%	28.15%	31.54% [19.67, 36.67]	35/4/6
Radio, $h/a_{\text{ref}} = 0.5$	Connected	1.59%	4.15%	5.59% [2.36, 6.45]	26/3/16

TABLE 6. Modern static comparison (mean objective / mean rank).

Algorithm	Weighted	Unweighted
EAA-static	418.97 / 2.02	7.68 / 4.51
EAAS-S4C-MAB-R	418.97 / 2.02	—
PBIG-R	401.77 / 2.76	10.30 / 2.46
GH-MWDDS+	394.47 / 3.23	9.77 / 3.30
FSS-2026-R	258.83 / 5.28	10.70 / 1.76
MC-CMSA-R	244.21 / 5.68	9.98 / 2.96

F. FAILURES AND COMPONENT ABLATION

At $h = 0.25$, CARE exceeds the per-instance dynamic oracle by 2.78%, 2.38%, and 2.13% under no, random, and low-reserve-node removal for ordinary domination. Connected gains are 3.39%, 3.27%, and 2.94%; all six are Holm-significant. EAAS degrades from 314.09 rounds without failures to 178.31 under random removal, whereas CARE rebuilds and obtains 486.64. Low-reserve removal produces a larger absolute survivor lifetime than no removal. This is not presented as beneficial failure: removing a vertex also removes its service demand, so the experiment measures continuity for the surviving universe. Random failure gives the less ambiguous robustness comparison.

Removing neighborhood reserve reduces mean lifetime in every ablation cell. At $h = 0.5$, the reduction is 1.71% for ordinary and 1.87% for connected domination. Removing switch awareness reduces lifetime by 1.27% and 1.39% and

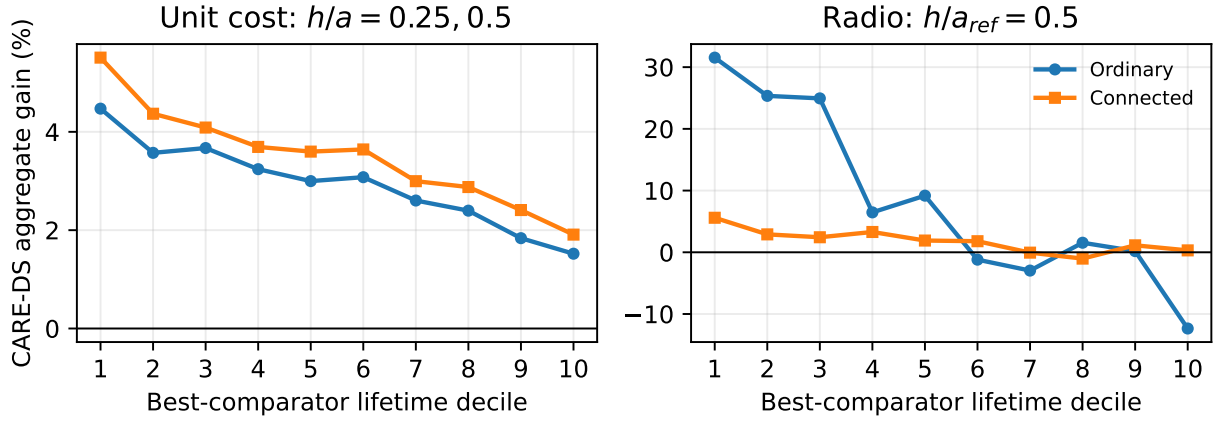
increases mean membership changes from 943 to 2,868 and from 940 to 3,080, respectively. The no-switch contrasts at $h = 0.1$ are not significant in either mode; the reserve contrast is significant in all six cells.

G. CONTROL-PARAMETER SENSITIVITY

Across the 100-graph sensitivity set, changing one control at a time leaves the qualitative result stable. Relative to the declared baseline, the mean lifetime change over the four mode-handover cells ranges from -1.20% to +0.89% for the tested dwell scales, from -0.44% to +0.98% for the tested refresh intervals, and from -0.64% to +0.34% for zero or eight randomized starts. Individual graphs can be more sensitive (the observed extremes are approximately -23.0% and +17.6%), so these parameters remain deployment controls rather than universal constants. More frequent refresh generally increases runtime; it does not create a systematic mean lifetime advantage large enough to explain the main result.

H. REPEATED ALGORITHM SEEDS

The ten-seed robustness subset confirms that the shared stochastic stream in the full matrix does not create the principal result (Table 7). Inference first averages the ten repetitions within each graph. CARE exceeds MS-RG-WT on all 40 ordinary graphs in both cells and on 36/40 and 37/40 connected graphs. Its mean within-graph coefficient of variation is 1.31–



1 = lowest competitor lifetime within each fixed design stratum; 10 = highest

FIGURE 2. Exploratory CARE-DS gain by within-stratum decile of the best observed comparator lifetime. Decile 1 is the lowest-lifetime tail and decile 10 the highest. Unit-cost results pool the two material-handover cells; radio uses the nonzero wake-up cell. Every point is a ratio of aggregate means.

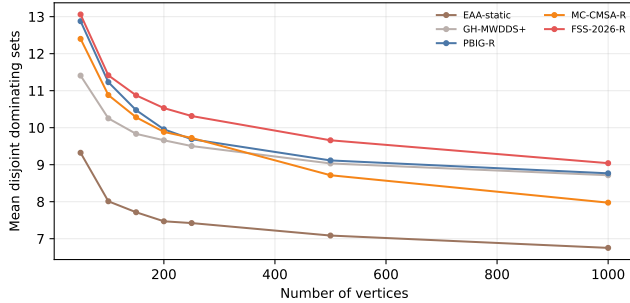


FIGURE 3. Unweighted MDDSP validation up to 1,000 vertices.

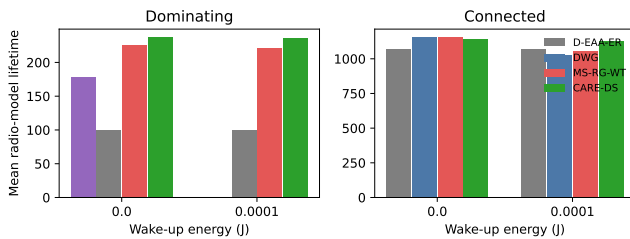


FIGURE 4. First-order radio lifetime. EAAS is defined only for ordinary domination.

2.02%.

All four graph-level Wilcoxon tests remain significant after Holm correction (largest adjusted $p < 2.0 \times 10^{-7}$); matched rank-biserial correlations are at least 0.991. On the 36-graph native-unweighted subset, FSS-2026-R remains strongest across ten seeds, with mean 9.98 sets, best or tied on 27 graphs, and 0.83% mean gap to the per-instance best. PBIG-R and MC-CMSA-R obtain means 9.62 and 9.15.

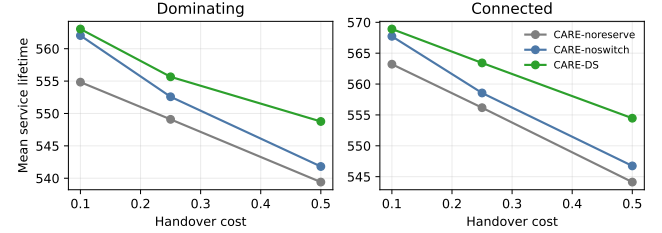


FIGURE 5. Submission-scale component ablation.

TABLE 7. Ten-seed dynamic robustness subset (40 graphs per row). Gain compares aggregate graph-level means; CV is the mean within-graph coefficient of variation.

Mode	h	Gain	W/T/L	CARE CV	MS CV
Ordinary	0.25	14.61%	40/0/0	1.31%	1.96%
Ordinary	0.50	26.03%	40/0/0	1.35%	3.37%
Connected	0.25	11.67%	36/3/1	1.88%	1.91%
Connected	0.50	22.55%	37/2/1	2.02%	2.53%

I. EXACT STRESS CASES

On the fixed, outcome-independent 21–22-vertex suite, CARE, D-EAA-ER, and the other methods almost always attain the optimum, confirming correctness but providing little separation. We therefore report a second, explicitly adversarial ordinary-domination challenge set: twelve 22-vertex cases fixed from a disjoint calibration pool by low heuristic-to-valid-bound utilization. It is not used for population inference. CARE is optimal on 5/12 challenge cases with mean ratio 0.964 and minimum 0.885; D-EAA-ER is 5/12 with mean 0.933, and DWG and MS-RG-WT are 3/12 with means 0.926. Exact enumeration contains up to 2,102 minimal dominating sets. On this deliberately selected stress set, CARE-DS obtains the best aggregate performance among the tested heuristics, while its remaining 11.5% worst-case gap identifies room for

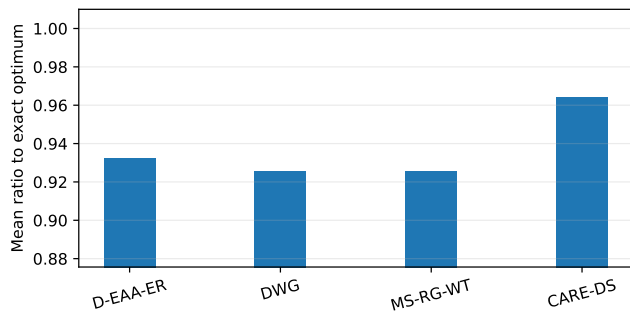


FIGURE 6. Ratio to optimum on the adversarial exact challenge set.

future matheuristic work. This stress-set result is not evidence of population-wide superiority.

X. DISCUSSION

A. INTERPRETATION OF THE MAIN RESULT

The experiments identify a boundary rather than an unconditional winner. When handover is free, a scheduler may change the active set whenever a better immediate energy balance is found. CARE-DS receives no compensation for its dwell rule in this case and need not lead. When entry costs are 0.25 or 0.5, frequent replacement begins to consume a substantial part of the same energy budget that supports service. The reserve coordinate and the dwell rule then address different parts of the problem: the former protects future coverage, while the latter avoids paying repeatedly to obtain small short-term gains.

On the sampled unit-cost grid, the aggregate CARE-DS comparison changes sign between $h/a = 0.10$ and 0.25 in both service modes. This interval is an empirical break-even region for the tested distributions, algorithms, and control settings; it is not a universal threshold or a claim of monotonic superiority on every topology.

The closed-neighborhood reserve also has a direct network interpretation. If a vertex v and all nodes in $N[v]$ are nearly depleted, no later choice elsewhere in the graph can restore service to v . Such a neighborhood is a local energy bottleneck. Maximizing total residual energy would not necessarily protect it, because energy in a distant dense region cannot be moved to $N[v]$. This is why the minimum local reserve is a more relevant signal than network-wide remaining energy.

The lower-tail result sharpens this interpretation. A fixed disjoint family can strand the residual energy of strong members when one weak member determines the usable duration of their set. CARE-DS can reconstruct an overlapping set and redirect work when alternative local dominators or relays exist. It is not independent of weak nodes: an unavoidable articulation-like relay or a depleted closed neighborhood remains a genuine service bottleneck. The evidence therefore supports reduced sensitivity to an accidentally weak member of a precomputed set, not immunity to structurally critical low-energy regions.

The connected experiments are more restrictive. A candidate must cover the live network and remain connected to the sink. BA and clustered graphs can have bridge-like regions through

which many candidate backbones must pass. The resulting lack of portfolio diversity explains why these families contain the weakest cells. It also shows why ordinary domination results should not be used as a proxy for a routing backbone.

B. IMPLICATIONS FOR A WSN DEPLOYMENT

CARE-DS requires four pieces of state at a refresh event: the live-node graph, residual node energies, the previously active set, and an estimate of the cost of entering service. The handover parameter should represent the energy spent on wake-up, synchronization, schedule dissemination, and route formation at the layer where CARE-DS is deployed. It should not be treated as a tuning constant detached from the hardware or protocol.

The CC2420 calculation in the experimental design is only an order-of-magnitude anchor. A deployment should measure the complete wake-up, synchronization, control-message, and route-installation transaction on its own platform and map that measured energy to h/a (or h/a_{ref} in the radio model) before selecting a dwell scale.

The refresh interval and the dwell period serve different purposes. Refreshing reconstructs the candidate portfolio from new state information. Dwelling controls how quickly the selected set may change while it remains feasible. A deployment with stable links and reliable energy estimates can use infrequent refreshes. A deployment exposed to failures or link changes should refresh on the corresponding events, even when the regular interval has not expired.

The failure results must be read with the definition of continued service. A failed sensor is removed from the selectable set and from the service universe. Consequently, removing a low-reserve node can increase the lifetime of the surviving network by removing a difficult service demand. This does not make failure beneficial. The random-failure comparison gives the cleaner evidence that reconstruction preserves service better than retaining a fixed disjoint family.

Static and dynamic methods answer different questions. FSS is designed to maximize the number of pairwise-disjoint sets and should first be judged on the unweighted MDDSP suite, where it is strongest. Heterogeneous lifetime also depends on the bottleneck energy within each set and on what residual energy is left outside the static family. CARE-DS is therefore not proposed as a replacement for offline MDDSP when the application genuinely has unit energy, zero transition cost, no failures, and a fixed family. In that regime the simpler static formulation is sufficient.

XI. THREATS TO VALIDITY

The study is simulation-based and centralized. Perfect topology and residual-energy knowledge are assumed at refresh events. The energy required to collect this state is not charged separately. The first-order radio model includes transmission, reception, aggregation, and wake-up energy, but it does not simulate interference, contention, retransmission, fading, clock synchronization, or the full control plane. The scalar handover cost summarizes several protocol actions that may vary across

devices. Accordingly, the positive result is supported only above the observed break-even region in the tested sweep and should not be extrapolated to systems with negligible entry cost. The powered sink assumption favors connected service. Failure events remove nodes cleanly rather than modeling intermittent links or correlated regional damage.

PBIG-R, MC-CMSA-R, FSS-2026-R, and EAAS-S4C-MAB-R are not authors' binaries. They are mechanism-audited reproductions, and the manuscript does not claim bit-for-bit equivalence. Equal wall time is approximate because a single Python construction or MILP call can finish slightly after its deadline; actual runtime is retained. Graph families and parameter ranges are broad but cannot represent every physical deployment. The results establish the behavior of the scheduling principle under controlled conditions; they are not a substitute for packet-level or hardware validation.

Each stochastic heuristic uses one deterministic algorithm seed per graph in the full matrix. Pairing controls graph variation; the separate ten-seed subset estimates within-instance variability on a stratified sample, but it is not a full repeated-run metaheuristic benchmark. Runtime-control conclusions apply to the declared 0.5-s Python regime, not to the longer budgets used in the original PBIG and FSS studies.

The lower-tail analysis is post hoc. Defining the tail through the best observed comparator can introduce selection and regression-to-the-mean effects, although ranking within fixed design strata prevents graph size, topology, density, or handover setting from manufacturing the pattern. The stratified bootstrap quantifies sampling uncertainty conditional on this design, but does not turn the analysis into confirmatory evidence. A held-out matrix or testbed study should prespecify the decile rule before treating lower-tail protection as a primary deployment claim.

XII. REPRODUCIBILITY

The artifact fixes all seeds and configurations, validates every returned set, supports resumable instance-level CSV output, and contains integrity checks for expected row counts, duplicate keys, missing values, and invalid schedules. Paper numbers and figures, including the stratified lower-tail analysis, are regenerated from raw results. The baseline audit states exactly which published mechanisms were reproduced and which details remain implementation choices.

XIII. CONCLUSION

We studied WSN lifetime as a sequence of energy-feasible dominating sets rather than as a fixed family of pairwise-disjoint sets. The formulation allows residual energy to be reused, charges newly activated sensors, reconstructs the schedule after failures, and optionally requires a sink-rooted connected backbone.

Every unit-cost service round must use energy from each closed neighborhood. This gives the proved resource upper bound. CARE-DS extends the same local bottleneck idea to the radio model as a routing-aware heuristic score; no radio optimality bound is claimed. The algorithm constructs a small

portfolio of feasible sets, selects the set that best preserves the minimum reserve, and limits unnecessary handovers by a dwell rule. The ablation confirms that both parts contribute to lifetime.

The experimental conclusion has a clear scope. CARE-DS is not best when changing the active set is free. It improves lifetime when handover energy is material, remains effective when failures require reconstruction, and shows its largest exploratory gains on instances where the competing schedules terminate early. The modern static, radio, failure, exact, and portfolio-controlled experiments support this claim without extending it beyond the model. Packet-level and testbed studies are the natural next step for deciding how accurately the scalar handover cost represents a deployed protocol.

DATA AND CODE AVAILABILITY

The supplementary research artifact contains the source code, fixed configurations, deterministic seeds, raw result tables, analysis scripts, validation checks, and figure-generation workflow used in this study. A public archival repository and persistent identifier may additionally be supplied during publication; all material needed to reproduce the reported tables and figures is included in the supplementary package submitted with this manuscript.

ACKNOWLEDGMENT

The authors conceived the research problem, mathematical model, CARE-DS algorithm, comparator protocol, experimental design, interpretation, and conclusions. OpenAI Codex was used under the authors' direction to implement their algorithmic specifications, prepare regression and validation tests, execute and check the statistical-analysis and table/figure-generation scripts, generate computational reports, assist with language editing, and prepare the IEEE Access LaTeX source [21]. Its use therefore affected the software artifact, the statistical outputs, tables, and figures reported in the Results section, and editorial wording throughout the manuscript. The authors inspected the implementation, reran the validation workflow, checked the numerical results and references, revised the generated text, and accept full responsibility for the accuracy, originality, and integrity of the work. No AI system is listed as an author or treated as a source of scientific evidence.

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